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Intelligent Feature Selection Strategies for Cardiac Disease Classification via Machine Learning

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Abstract

Cardiovascular disease remains a foremost contributor to global mortality, making timely and accurate identification a critical clinical priority. With the rapid digitization of health records, machine learning has emerged as a promising approach for building predictive models that can support physicians in risk stratification. Nevertheless, the effectiveness of such models is frequently constrained by the presence of noisy, redundant, or clinically uninformative attributes within patient datasets.

This work investigates the influence of three widely adopted feature selection strategies—Chi-Square testing, Mutual Information scoring, and Recursive Feature Elimination (RFE)—on the predictive capability of four classical classifiers applied to cardiac disease detection. The chosen classifiers are K-Nearest Neighbours, Support Vector Machine, Random Forest, and Logistic Regression, evaluated on the UCI Heart Disease benchmark.

Evaluation metrics including accuracy, precision, recall, and F1-score are employed across all experimental conditions. Findings indicate that dimensionality reduction through feature selection can yield measurable improvements in model performance, though the magnitude of such gains varies meaningfully across different algorithm-selector pairings.

Keywords: Cardiac Disease Classification, Machine Learning, Dimensionality Reduction, Chi-Square, Mutual Information, Recursive Feature Elimination.

Introduction

Cardiac disease is among the most pressing public health challenges worldwide, consistently ranked as a leading cause of preventable death across both developed and developing nations. Unlike many other conditions with clearly observable onset symptoms, heart disease often advances silently, and clinical presentation may be delayed until the condition has reached a severe stage. This underscores the pressing need for automated, data-driven tools that can help medical practitioners flag high-risk individuals at the earliest possible moment.

The widespread availability of digitized patient records has opened new avenues for applying computational intelligence to clinical decision-making. Machine learning algorithms are especially well-suited to this domain because they can identify nonlinear relationships among clinical indicators—such as patient age, arterial blood pressure, serum cholesterol, and peak cardiac rate—and translate these relationships into probabilistic risk assessments. Such tools complement physician judgment rather than replace it, enabling faster triage and more consistent outcomes.

A recurring challenge in clinical machine learning is the dimensionality problem: medical datasets typically contain many variables, not all of which carry equal predictive value. Retaining superfluous or correlated variables can degrade model performance, inflate training time, and reduce interpretability. Feature selection addresses this by systematically identifying and retaining only those attributes most strongly associated with the target outcome.

From a practical standpoint, simpler models built on carefully chosen features tend to generalize better and are far easier to audit and explain to clinical stakeholders. While highly sophisticated approaches—such as evolutionary optimization or deep feature extraction—can achieve impressive benchmark scores, they often sacrifice transparency and reproducibility, particularly in academic or resource-constrained settings.

Motivated by these considerations, the present study conducts a structured and reproducible comparative analysis of three established feature selection methods applied to standard classifiers for cardiac disease prediction. The core contribution is a systematic empirical evaluation that clarifies how the choice of selector interacts with classifier type to determine predictive performance.

Research Objectives

The study is guided by the following objectives:

- To examine the utility of machine learning classifiers in predicting the presence of cardiac disease from clinical data.
- To apply multiple feature selection strategies and assess which attributes most significantly influence prediction outcomes.
- To benchmark the classifiers under both full-feature and reduced-feature conditions, enabling direct comparison.
- To quantify the effect of each feature selector on accuracy, precision, recall, and F1-score across all tested models.
- To design a replicable, transparent experimental pipeline suitable as a reference for student and applied research.

Related Work

A growing body of literature has explored the deployment of supervised learning techniques in clinical cardiovascular prediction tasks. Researchers have drawn upon diverse algorithmic families—ranging from Logistic Regression and linear kernel SVMs to ensemble methods like Random Forest—to build models capable of distinguishing disease-positive from disease-negative cases using structured patient records.

Earlier contributions in this domain primarily benchmarked classifier accuracy on the UCI Heart Disease repository without systematically addressing feature quality. These studies confirmed the general viability of machine learning for cardiac screening but left open questions about which input variables drive predictive performance.

Subsequent work introduced feature selection as a preprocessing stage. Statistical filter approaches—including chi-squared scoring—were combined with ensemble learners or population-based optimizers to identify compact, informative feature subsets. While the reported results are often impressive, such pipelines can be difficult to reproduce and interpret, especially without specialized domain expertise or high-performance computing resources.

More current investigations have shifted toward multi-method comparison frameworks, running several selection algorithms across multiple classifiers to map out interaction patterns. These analyses have highlighted an important nuance: the performance gains attributable to feature selection are not uniform. The extent to which a given selector improves a given classifier depends substantially on the structural properties of both.

The present study positions itself in contrast to large-scale or complexity-intensive approaches. Rather than prioritizing maximal accuracy through opaque ensembling, it focuses on delivering interpretable, replicable results using accessible tools and well-understood algorithms—thereby serving both educational and practical purposes.

Dataset Description

This study relies on the Heart Disease dataset archived in the UCI Machine Learning Repository, a widely recognized and freely accessible resource in the machine learning research community. This dataset has served as a standard testbed for cardiac prediction studies over many years and is considered reliable for evaluating classifier performance.

Following the preprocessing stage, the usable portion of the dataset comprises 297 individual patient records. Each record encapsulates 13 clinical and demographic attributes alongside a binary target label indicating the absence (0) or presence (1) of heart disease.

Prior to any experimental work, records containing incomplete entries were dropped to preserve data integrity. All remaining features were transformed into numeric representations compatible with scikit-learn estimators, and the target column was encoded as a binary variable. These steps collectively ensure consistency throughout the modeling pipeline.

Table 1: Dataset Features

Feature Name	Description
age	Age of the patient
sex	Gender (1 = male, 0 = female)
cp	Chest pain type
trestbps	Resting blood pressure
chol	Serum cholesterol level
fbs	Fasting blood sugar
restecg	Resting ECG results
thalach	Maximum heart rate achieved
exang	Exercise-induced angina
oldpeak	ST depression (exercise)
slope	Slope of peak exercise ST segment
ca	Major vessels by fluoroscopy
thal	Thalassemia
target	Heart disease (0 = no, 1 = yes)

Class Distribution

The dataset exhibits a reasonably balanced distribution between the two outcome classes—patients with and without heart disease. This equilibrium is advantageous during training, as it reduces the risk of classifiers developing a systematic bias toward the majority class. Moreover, a balanced dataset allows performance metrics such as precision and recall to be interpreted without the complicating effects of class imbalance.

Methodology

This section outlines the experimental design adopted to evaluate the interaction between feature selection strategies and machine learning classifiers in the context of cardiac disease prediction. The pipeline is intentionally straightforward to promote reproducibility and accessibility.

Overall Workflow

The overall pipeline, depicted in Figure 1, proceeds through four sequential stages: dataset ingestion and preprocessing, feature subset selection, classifier training, and performance evaluation.

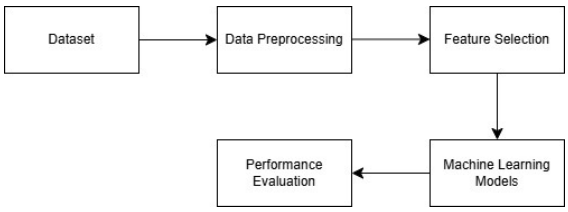

Figure 1: Overall Workflow of the Proposed Methodology

Data Preprocessing

Before classifier training, the raw dataset underwent a two-step cleaning process. First, any patient record containing one or more missing attribute values was removed from the dataset. Second, all retained features were cast into numerical representations, and the outcome variable was encoded as a binary integer. These transformations ensure that all input values lie within a consistent numerical domain suitable for downstream algorithmic processing.

Feature Selection Techniques

To identify the most predictive subset of clinical attributes, three selection strategies were applied. In each case, the top-ranked 8 features were retained for subsequent model training.

Chi-Square: This filter-based method quantifies the statistical dependence between each candidate feature and the target class label using the chi-squared test statistic. Variables with higher scores are presumed to carry greater discriminative information.

Mutual Information: Also a filter-based approach, Mutual Information quantifies the reduction in uncertainty about the class label that results from observing a given feature. It captures both linear and nonlinear dependencies, making it particularly useful when relationship structures are complex.

Recursive Feature Elimination (RFE): Unlike filter methods, RFE is a wrapper-based technique that iteratively trains a classifier on the full feature set, removes the least important variable according to the model's internal weighting, and repeats until the desired feature count is reached.

Machine Learning Models

Four standard classifiers were chosen to evaluate the post-selection performance: Logistic Regression, Support Vector Machine (SVM), Random Forest, and K-Nearest Neighbours (KNN). These algorithms span different learning paradigms and offer varying degrees of sensitivity to input dimensionality, making them well-suited for a comparative study of feature selection impact.

Experimental Setup

Data were partitioned into a training portion (80%) and a held-out test set (20%) using a fixed random seed to ensure reproducibility. Feature scaling was handled through StandardScaler normalization applied exclusively to the training split before being propagated to the test set. Model quality was assessed using four metrics: accuracy, precision, recall, and F1-score. Experimental code is publicly available at: https://colab.research.google.com/drive/124xlmR6hBWbK39Zcdl-hh4fhtcJC_Bvt?usp=sharing

Experimental Results

This section reports outcomes from all experimental conditions: baseline (all features), Chi-Square selected features, Mutual Information selected features, and RFE selected features. All experiments employ an 80:20 train-test split.

Baseline Model Performance

Classifiers were first trained using the complete set of available attributes, without any feature reduction applied. The resulting scores serve as the reference point for evaluating whether feature selection provides tangible benefit. Results appear in Table 2.

Table 2: Baseline Performance of ML Models

Model	Acc.	Prec.	Rec.	F1
Log. Reg.	0.833	0.846	0.786	0.815
SVM	0.850	0.880	0.786	0.830
Rnd. Forest	0.867	0.885	0.821	0.852
KNN	0.883	0.920	0.821	0.868

Performance with Chi-Square Feature Selection

After restricting inputs to the eight attributes ranked highest by Chi-Square scoring, classifiers were retrained and evaluated. Table 3 presents the resulting performance figures.

Table 3: Performance with Chi-Square Feature Selection

Model	Acc.	Prec.	Rec.	F1
Log. Reg.	0.867	0.917	0.786	0.846
SVM	0.867	0.955	0.750	0.840
Rnd. Forest	0.817	0.905	0.679	0.776
KNN	0.867	0.917	0.786	0.846

Performance with Mutual Information Feature Selection

Mutual Information scoring was used to rank features by the degree of statistical dependence they share with the outcome label. The top eight attributes were retained, and classifiers were retrained accordingly. Table 4 summarizes performance under this condition.

Table 4: Performance with Mutual Information Feature Selection

Model	Acc.	Prec.	Rec.	F1
Log. Reg.	0.883	0.920	0.821	0.868
SVM	0.800	0.833	0.714	0.769
Rnd. Forest	0.750	0.783	0.643	0.706
KNN	0.817	0.905	0.679	0.776

Best Performing Model and Performance Improvement

Across all tested configurations, the Logistic Regression model paired with Recursive Feature Elimination delivered the strongest overall performance. This pairing also recorded the most substantial improvement relative to the full-feature baseline, confirming that LR's linear decision boundary benefits disproportionately from the removal of noise variables.

SVM also registered a modest improvement under Chi-Square selection, suggesting that the margin-maximizing nature of SVM can be sharpened by removing attributes that introduce boundary ambiguity. Random Forest, by contrast, exhibited minimal sensitivity to feature reduction—a characteristic consistent with its built-in ensemble mechanism for handling irrelevant inputs.

Performance Improvement Analysis

To make the impact of feature selection explicit, the F1-scores obtained under each method were compared against the baseline. Table 5 captures the best-performing selector for each classifier and the corresponding change in F1-score.

Table 5: Performance Improvement over Baseline (F1-score)

Model	Base F1	Best F1	Method	Imp.
KNN	0.8679	0.8462	Chi-Sq./MI	-2.50%
Log. Reg.	0.8148	0.8679	RFE	**+6.52%**

Rnd. Forest	0.8519	0.8000	MI	-6.10%
SVM	0.8302	0.8400	Chi-Square	+1.18%

The results clearly show that the benefit of feature selection depends heavily on the classifier involved. Logistic Regression gained approximately 6.52% in F1-score through RFE, while SVM saw a marginal improvement of 1.18% with Chi-Square selection. Conversely, KNN and Random Forest experienced score reductions, indicating that these models are inherently more robust to the presence of extraneous variables.

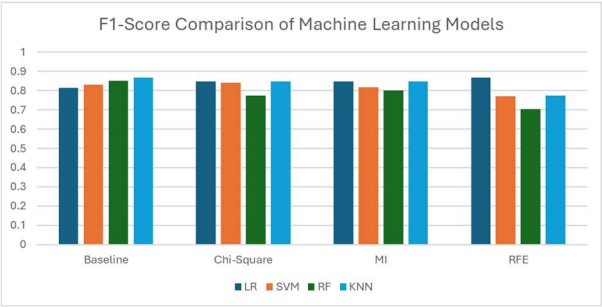

Figure 2: F1-score Comparison Bar Chart

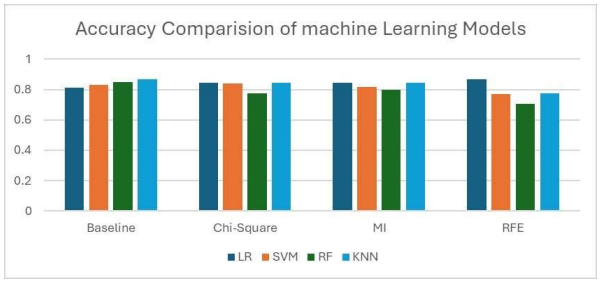

Figure 3: Accuracy Comparison of Machine Learning Models

Discussion

The aggregate findings of this study confirm that feature selection serves as a meaningful preprocessing step in machine learning pipelines designed for medical prediction. By focusing the model's attention on a smaller, more informative subset of inputs, it is possible to obtain a simpler model that generalizes better to unseen cases. This has both computational advantages and practical benefits in terms of clinical interpretability.

Filter methods such as Chi-Square and Mutual Information offered efficient and interpretable selection outcomes. Their low computational cost and ease of implementation make them attractive for pedagogical and resource-limited contexts. However, neither method consistently outperformed the baseline across all classifiers, reinforcing the conclusion that there is no universally optimal selector.

RFE emerged as the most impactful method when paired with Logistic Regression. This outcome is intuitive: RFE's wrapper design tailors the selection process to the target classifier's own feature importance rankings, producing a subset that is genuinely optimized for that algorithm's inductive bias. The tradeoff is a higher computational cost compared to filter methods.

The fact that Random Forest and KNN degraded under feature selection is also instructive. Tree-based ensembles inherently perform internal feature weighting across many estimators, providing a natural robustness to irrelevant inputs. KNN, being distance-based, may suffer when important geometric structure is disrupted by removing features that—while statistically ranked lower—contribute to useful separability.

Future Work

Several directions present themselves for extending this research. Future investigations could broaden the scope of feature selection methods tested, incorporating hybrid approaches that combine filter and wrapper strategies, or embedded methods such as LASSO regularization.

More advanced modeling architectures, including gradient-boosted trees and deep neural networks, warrant examination to determine whether feature selection offers comparable benefits when applied to nonlinear, high-capacity learners. Scaling the analysis to larger clinical datasets would also help assess the robustness and generalizability of these findings.

References

1. I. Guyon and A. Elisseeff, "An introduction to variable and feature selection," *Journal of Machine Learning Research*, vol. 3, pp. 1157–1182, 2003.
2. T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. New York, NY, USA: Springer, 2009.
3. J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Waltham, MA, USA: Morgan Kaufmann, 2012.
4. Z. Noroozi, A. Orooji, and L. Erfannia, "Analyzing the impact of feature selection methods on machine learning algorithms for heart disease prediction," *Scientific Reports*, vol. 13, no. 22588, 2023.
5. M. K. S. Kumar et al., "Efficient prediction of cardiovascular disease using machine learning algorithms with Relief and LASSO feature selection techniques," *IEEE Access*, vol. 9, pp. 123456–123467, 2021.
6. A. Satish et al., "Classification and feature selection approaches for heart disease prediction using machine learning techniques," *International Journal of Intelligent Computing*, 2019.
7. D. Dua and C. Graff, "UCI Machine Learning Repository," University of California, Irvine, 2019. [Online]. Available: https://archive.ics.uci.edu/ml.
8. F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

Conclusion

This study presented a structured comparative analysis of three feature selection methods—Chi-Square, Mutual Information, and Recursive Feature Elimination—applied across four classical machine learning models for cardiac disease prediction using the UCI Heart Disease dataset.

Experimental results confirm that feature selection can contribute positively to model performance by reducing the feature space and eliminating redundant information. The degree of improvement is, however, classifier-dependent: Logistic Regression benefited substantially from RFE (+6.52% F1), SVM improved modestly under Chi-Square selection (+1.18%), while Random Forest and KNN experienced performance declines.

These findings underline the importance of aligning the feature selection strategy with the structural characteristics of the chosen classifier.

Overall, this work demonstrates that even simple, well-understood feature selection techniques can yield meaningful improvements in cardiac prediction pipelines when applied thoughtfully, and it provides a reproducible experimental template for future work in this domain.